

# Referee Reports — WR10260/Suarez

## Report of the First Referee — WR10260/Suarez

The authors investigated the social network structure around three social movements resulting in demonstrations. They claim to find a critical transition and an increase in self-similarity to be tied to critical time points.

While the topic, approach and findings are potentially interesting and suited for Physical Review Research, the authors do not explain their methodology sufficiently and failed to convince regarding their central claims (details below). I therefore recommend a substantial revision regarding these points before the manuscript can be reconsidered for publication.

### Methodology

The description and motivation is lacking to the point where I could not understand certain parts. There is a strong verbosity dissonance: Some things such as nestedness and the Kronecker delta are described and motivated in more detail than necessary, while the reader is just expected to be familiar with other things, even though uncommon. I give some examples for this below, but the authors should revisit all descriptions and motivations of methods to ensure completeness and intelligibility by an audience not intimately familiar with the authors' previous works.

- “Based on this exploration, tweets related to the 9n mobilisation were aggregated into one-hour windows, whereas those associated with the No al Tarifazo and Charlie Hebdo movements were aggregated into two-hour windows.” – This seems rather arbitrary, in particular the apparent restrictions to windows that are a multiple of an hour. Can the authors undermine their choices by minimising some metric or similar?

Re: We set a minimum threshold of 700 unique hashtags to define the window width. (Ver nuevo párrafo azul para decidir si quitamos el anterior), **yo entropía**.

- The explanation of the modularity is lacking: I need to guess that  $g_i$  and  $g_j$  encode the groups. Is  $Q$  minimised over groupings?

Re: It's right. Corrected.

- The metric  $\epsilon^2$  is very badly described. The main manuscript doesn't elaborate it at all and the appendix refers to some bins that aren't introduced. Ref. 26 is suffering from the same problem. On top, the authors call it a “test”, but it does not appear to be one.

We thank the Referee for encouraging us to describe our methods more precisely. We have revised the manuscript to provide a clear explanation of the collapse-quality metric  $\epsilon^2$ , which was previously introduced in Ref. 26. The Referee is right that  $\epsilon^2$  is not a statistical test, and we have therefore removed this terminology from the manuscript.

- The authors often use the word “collapse”, but do not specify what they mean. In some cases it seems to be a “sudden drop” of an observable; in others it seems to be observables aligning for multiple datasets. Very often it is also unclear what (observable) is collapsing.

Re: Corrected.

- How does modularity change over time when groups are fixed? This should give us an idea whether it’s predominately the groups or the connections that are changing.

Manuel: Test a pares de hashtags ([https://en.wikipedia.org/wiki/Adjusted\\_mutual\\_information](https://en.wikipedia.org/wiki/Adjusted_mutual_information)).

- “On the other hand, as the division in TW’s is a way to discretise the continuous time range of the timespan the dataset covers, it can cause anomalies in certain TWs. To avoid these noise errors, we decided to calculate the moving average of the last  $N = 3$  TW’s.” – Why not use overlapping time windows for the moving average to get a better sampling while keeping a fine time resolution? Also, if removing outliers is the goal, why not use the median instead of the mean?

Hacer un test a ver si queda mejor con overlapping time windows y explicar que no lo hacemos para quitar outliers sino para suavizar fluctuaciones.

## Argumentation

The main claims of the paper are not very well substantiated. Most importantly:

- The authors do not consider alternative explanations and in particular not any null models. In particular I would be interested whether simpler and established properties of the networks can explain the authors’ observations.

The fact that the networks present high levels of clustering and DTR self-similarity at the CTWs reduces the portfolio of models that can be used to explain the observations. On the other hand, one would like the models to be maximum entropy models, such that networks in the corresponding ensemble are maximally random except for the assumed constraints. As far as we know, the geometric  $\mathbb{S}^D/\mathbb{H}^{D+1}$  is the only family of models that produce networks with high levels of clustering and DTR self-similarity while being maximally random ensembles, as proved in Phys. Rev. Lett. 135, 197402 (2025) and arXiv:2603.18170. We have added a clarification to the discussion section.

- The reasoning is somewhat circular, as parameters are optimised to show what is later claimed as a finding, specifically a sharp modular-to-nested transition and a sharp drop of  $\epsilon^2$  (for 9n).

This interpretation is not correct. We do not optimize any parameters; rather, we perform measurements on structured data within the TWs. However, we acknowledge that the previous wording of the manuscript may have conveyed this impression because our description was not sufficiently precise. In the revised version, we have clarified the methodology.

- The work is based on only three events, two of which happened in the same country. How were these chosen? Why only these three? I would imagine that many similar events can be identified with specific hashtags and analysed using the same pipeline without much effort.

Re: We thank the referee for this question. Our choice of events was guided first by a methodological requirement and second by a practical constraint on data access. Methodologically, the three events were selected to allow a homogeneous and controlled data collection under a single, consistent protocol. They also span two countries (Argentina and France), two languages (Spanish and French) and two distinct types of mobilisation: two episodes of socio-economic

and institutional protest in Argentina (No al Tarifazo and 9N) and a reactive solidarity mobilisation in France (Charlie Hebdo). Observing the same self-similar behaviour across different national, linguistic and thematic contexts strengthens, rather than weakens, the claim that the phenomenon is not an artefact of a single setting.

The referee is correct that the analytical pipeline itself is general and can in principle be applied to many hashtag-identifiable events with little additional effort. The limiting factor is not the analysis but access to comparable, well-controlled data. The datasets used here were purchased through a research grant obtained in 2022, which provided dedicated funding for data acquisition. Data from X (formerly Twitter) are costly. The academic access routes that previously allowed researchers to collect such data at scale were discontinued under the platform's current ownership. Alternative platforms do not yet reach a comparable level of adoption, and X remains the medium where users concentrate during large-scale, real-time events. For these reasons, expanding the corpus under the same homogeneous collection conditions is currently constrained by data availability rather than by the method.

Several claims are based on eyeballing instead of a more quantified approach.  
Some specific issues related to this:

- “Notably, the identified CTWs align with the moments when digital activity translated into massive physical demonstrations, suggesting that the underlying network structures reflect fundamental organisational principles of collective attention” – Can the authors mark the times of these demonstrations in Fig. 2 and similar? Unless I am mistaken, the critical time windows for at least 9n and Charlie Hebdo coincide with the respective demonstrations. Could it be that what the authors are seeing is not a mobilisation event but simply people being the impact of the actual demonstrations on posting behaviour, e.g., people being at the demonstrations only posting about this but not reacting anymore or engaging in other topics as they are busy demonstrating (and posting about it). If the special time points found by the authors relate to the organisation of the demonstration, they need to happen some time before the actual demonstrations – somebody who is more familiar with these things can probably say how much “some time” is.

Re: We thank the referee for this point, as it lets us correct a possible misreading and state our claim precisely. Our claim concerns collective attention, not causality. We do not assert that the network signature organises, causes, predicts or precedes the demonstrations. The expression "organisational principles of collective attention" refers to how collective attention itself becomes structured, that is, the modular-to-nested reorganisation and the self-similar behaviour that emerge in the co-occurrence network when attention concentrates on a single unfolding event. It does not refer to the logistical organisation of the demonstration. We believe the referee's concern follows from reading "organisational" in this second sense, and the requirement that the signature must precede the demonstration applies only under that reading, we apologize for the misunderstanding.

Once the claim is read as a statement about attention, the alternative the referee proposes is not in competition with it. Whether a user is physically present at the demonstration and posting from it, or following and engaging with the event remotely, both are directing sustained attention to the same situation, and both contribute to the same structural signature. This is precisely the phenomenon we characterise. Our analysis does not separate "organising" from "reacting", and

it does not need to, because the object of study is the structure of concentrated attention, not the behaviour or timing of the physical event. For the same reason we prefer not to mark the times of the demonstrations in Fig. 2 and the analogous figures. The object of analysis is the temporal window over which attention concentrates, not the instant of the demonstration. Marking that instant would suggest a temporal correspondence we do not claim, and demonstrations have no sharply defined moment in any case, since some participants can arrive earlier and leave later.

To avoid this misreading by future readers, we have added a clarifying sentence to the section Discussion of the new version of the manuscript making explicit that "organisational principles of collective attention" refers to the self-organisation of attention in the network structure, and not to the logistical organisation of the demonstrations, and that we make no causal or predictive claim regarding the physical events.

- “Our interpretation is that this scale-invariant behaviour emerges from the coordinated efforts of participants to achieve common goals within compressed timeframes.” – This requires some more explanation.

Vamos a esperar el resultado del test del path lenght o la versión de la difusión. Si el camino es más corto que los no críticos → podemos reescribir hacia "auto-organización que reduce la distancia topológica media respecto a lo esperado por azar, consistente con una estructura que favorece la difusión eficiente del mensaje". Hay que coger los embedding que tenemos y calcular la navigability que son el strength y success. Básicamente, consiste en cuantificar cuantos paquetes llegan de un lado al otro y lo otro es el camino más corto. Mariángeles nos puede facilitar un código pero dice que se encuentra fácilmente en la net.

- The entire section on hyperbolic embeddings is detached from the rest of the manuscript. What is the connection to the social dynamics or to the critical transition points claimed by the authors?

Our claim, supported by the quantitative results presented in the manuscript, is that online social interactions, represented by hashtag co-occurrence networks generated from users’ activity on online platforms, exhibit signatures of multiscale  $k$ -cut self-similarity during periods of peak mobilization across diverse geographical and sociopolitical contexts. Together with the high levels of clustering observed in these networks, this evidence strongly suggests the presence of an underlying geometric structure that accounts for the highly organized interaction patterns.

In particular, the family of geometric network models in hyperbolic space has been shown to display this property in previous work (see Refs. xxx and yyy in the revised manuscript) and to provide an excellent quantitative explanation for the main structural features and symmetries of real-world networks. We therefore adopt this modeling framework and use hyperbolic embeddings to obtain a geometric representation of the hashtag co-occurrence networks, thereby defining effective distances between hashtags.

In the revised manuscript, we further show that greedy routing on these embeddings achieves high navigability during critical mobilization times indicating enhanced communicability as compared with non-critical TWs. TO BE DONE.

- “Despite their geographical and sociopolitical diversity” – For those readers who are not super familiar with the younger Argentinian history, can the authors provide some information on how diverse the two Argentinian events were?  
It’s now expanded in the section DATASETS.

## Other medium and minor issues

- Consistently name Twitter. I would suggest just “Twitter” as this was the name of the platform when these events happened and there will probably only ever be a small population who knows X but not Twitter. Given that this paper should also be read decades into the future, I also suggest to briefly characterise the nature of this platform (at the time when the data was taken).

Re: We agree with the referee and have standardised the wording in the discussion so that ‘Twitter (now X)’ appears, as in the other two instances where it was already written that way.

- “[self-similarity] lays the groundwork for the rapid amplification of information across scales that is necessary for the successful coordination of collective action.” – While I understand that this is not the main focus of the manuscript, the authors should briefly elaborate on potential mechanisms of this (not in the abstract).

We thank the Referee for drawing our attention to this passage and for helping us recognize that the original claim did not accurately reflect our results. What we mean is that the combination of self-similarity and high clustering provides strong evidence for an underlying metric space and therefore motivates the use of hyperbolic embeddings to construct a geometric representation of the hashtag co-occurrence networks, in which effective distances between hashtags can be defined. On these hyperbolic maps, greedy routing exhibits high navigability during periods of critical mobilization, indicating enhanced communicability compared with non-critical time windows. We have revised the text accordingly to make this point more precise and accurate.

- “However, the shortest path length approach has serious limitations in networks due to the small-world property, which limits the range of available shortest path-length scales [...]” – This claim requires more back-up.

We have provided more context for this claim.

- The outline paragraph (“The remainder of this paper ...”) is unnecessary, as the structure of this paper is pretty straightforward and the reader wouldn’t benefit from a roadmap (which is completely fine).

Re: We prefer to leave it in, as the ‘Methods’ section comes at the end. We think it costs nothing to include a guide to the structure of the manuscript.

- The authors should consistently use either  $k_t$  or  $k_T$ .

Re: Thanks for this remark. It’s corrected now.

- “ $n_{bins}$ ”  $\rightarrow$  “ $n_{bins}$ ” and similar.

Re: Thanks for this remark. It’s corrected now.

- “Following this procedure was easy to find the right TW” – This sentence is missing a word or two, but I am not sure which.

Re: Thanks for this remark. It’s corrected now.

- “Table ??”

Re: It’s corrected now.

- Fig. 2: Make the lines thicker so colours are easier to distinguish. Also there is no “black segmented” line, only a green dashed-dotted line. Finally, increase the scale for nestedness to make it better visible.

Re: We thanks the referee for those comments, which have been set. Manuel hay que poner la línea verde más gruesa y poner escala de nestedness a la derecha, de manera qu tenga sentido para que no quede espacio en blanco.

- I recommend that the authors consistently use local time to avoid explaining their conversion for some. I see no advantage of using UTC in the manuscript.

Manuel

- “Although the curves span a limited range of values, their collapse is clearly evident” – What curves does this refer to, what is a collapse here, and how is this remarkable?

Re: We thank the referee for these questions, which we address in turn.

*Which curves.* They are the complementary cumulative degree distributions (CCDFs) of the successive DTR subgraphs at the CTW, one curve per degree threshold  $k_t$ , with the degree of each subgraph rescaled by its own average degree,  $k_{\text{res}} = k/\langle k \rangle$ . This is stated in the same paragraph (Sec. III.C) and in the caption of Fig. 3.

*What “collapse” means.* The term denotes the extent to which the rescaled distributions of the renormalised subgraphs coincide once expressed in  $k_{\text{res}}$ , and it is quantified by the  $\epsilon^2$  test, where a low  $\epsilon^2$  is its operational meaning. The term was already used in Sec. III.A and in Methods B. To remove any ambiguity, in the revised version we now state this definition explicitly at the point where the term is first used. The visual collapse in Fig. 3a–c corresponds to the low  $\epsilon^2$  values at the CTW shown in Fig. 2g–i.

*Why it is remarkable.* Each DTR subgraph is a structurally distinct network of different size, density and composition, since at every step the least connected hashtags are removed. Preservation of the rescaled degree distribution across these renormalisation steps is therefore not automatic. Moreover, the property is specific to the CTW: at non-critical windows the same curves fail to collapse, with substantially higher  $\epsilon^2$  values (Fig. SI.3). The collapse also holds for a second, independent observable, the clustering spectrum (Fig. 3d–f). It is this convergence of two distinct observables at low  $\epsilon^2$ , rather than a visual impression over the limited range, that we regard as the robust result.

Apart from the explicit definition of collapse described above, no other changes were made to the manuscript in response to this comment.

- Fig SI3: Why not report a distribution of  $\epsilon_{cco}^2$  over all time points instead of a representative time point?

Re: The distribution of  $\epsilon_{cco}^2$  over all time points is in fact already reported. Fig. SI.2 shows the raw  $\epsilon_{cco}^2$  values computed for every temporal window across the three datasets, and Fig. 2 of the main text shows the same series after moving-average smoothing. The values quoted in connection with Fig. SI.3 are point values at the CTW and at a single non-critical window, provided only to illustrate the degree of curve collapse behind the CCDF and clustering-spectrum panels of that figure. To make this link explicit, we have added a cross-reference in Sec. III of the Supplementary Information pointing to the full  $\epsilon_{cco}^2$  series in Fig. SI.2 (raw) and in Fig. 2 of the main text.

- “From top to bottom: No al Tarifazo, 9N, and Charlie Hebdo datasets.” – Why clutter the caption with this when it is clearly visible in the actual plots (as it should be)? Captions should describe what the legend and labels cannot.

A discutir....A mi me gusta como está.

- “TW’s” → “TWs” (when it’s a plural and not a genetive.)

Re: Corrected. Both occurrences of “TW’s” in the manuscript were plurals and have been changed to “TWs”, consistent with the abbreviation as introduced in the text.

## Report of the Second Referee — WR10260/Suarez

This study investigates the structures and geometry of online networks during mass social mobilization events. Using three datasets from X and constructing user–hashtag and hashtag–hashtag co-occurrence networks, the authors find that the emergence of self-similar structures coincides with physical demonstrations. They also show that these networks during critical times can be embedded in hyperbolic spaces.

Overall, the study was reasonably executed, and the results are quite interesting, and I think it has potential for publication in PRR. However, there are some concerns about the data and interpretation of the results, and the presentation of the material needs serious improvement throughout and must be more thorough. Therefore, I would not recommend the paper for publication unless these issues are resolved or addressed convincingly.

### Major comments

- Their study is based on three datasets from X surrounding major mass mobilization events. Nonetheless, there is no literature or sources of information provided in the Datasets section to corroborate the claim that they all turned into street protests. In fact, I had a hard time identifying the “9n” event from external sources. Some sources also suggest that the “Charlie Hebdo” protests occurred across the globe and were not limited to France. Failing to provide these backgrounds can distort the understanding of the results presented and the data used. The authors should provide independent sources, such as from news outlets, that unambiguously identify each of these events including the time and whereabouts, and show that they are all well-documented protest events.

Re: We have expanded the explanation of the three demonstrations and added the corresponding reference. On the other hand, Tweets for the two Argentine mobilisations were restricted to those originating from Argentina, whereas tweets for the Charlie Hebdo dataset were restricted to those originating from France. We have added that quote to the manuscript.

- The global nature of the Charlie Hebdo mobilization implies that these hashtags shown in Table 1 may not necessarily be associated with the protest in France. Did the authors implement extra measures to account for such possibilities when gathering data?

Re: Yes, and this is now explained in the manuscript, as mentioned in the previous point.

- Some of the results overlap with Ref [28], including Fig. 2 (a–f). The authors should mention that these were already reported in the previous paper and highlight the shared patterns and possible differences, especially since they target the exact same events and data.

Re: We thank the referee for this observation. We confirm that panels (a–c) of Fig. 2 recompute activity statistics also reported in our earlier work [Rico and Gandica, Entropy, 2024], for the same three events. Panels (d–f), by contrast, report the same type of analysis (modularity and nestedness) but applied to a different network: the bipartite user–hashtag network, rather than the unipartite hashtag co-occurrence network used in that previous study. We have added a clarifying sentence to this effect in the revised manuscript.

- One of the paper’s main conclusions is that self-similar structures emerge when physical mass demonstrations were organized. Is this sign of criticality a predictor/precursor to or the outcome of these mass mobilization events? The authors note, “Our interpretation is that this scale-invariant behaviour emerges from the coordinated efforts of participants to achieve common goals within compressed timeframes,” implying that self-similarity in online networks reflects

coordinated behaviors driving mass demonstrations. However, this is not very convincing. It could very much be the outcome of these events: large-scale street demonstrations can garner significant attention from the public and increase user engagement in a particular way, leading to those nontrivial network organizations. Therefore, the causal relationships between these events and network characteristics should be probed more carefully or treated more cautiously.

Re: We thank the referee for this point, as it lets us correct a possible misreading and state our claim precisely. Our claim concerns collective attention, not causality. We do not assert that the network signature organises, causes, predicts or precedes the demonstrations. The expression "organisational principles of collective attention" refers to how collective attention itself becomes structured, that is, the modular-to-nested reorganisation and the self-similar behaviour that emerge in the co-occurrence network when attention concentrates on a single unfolding event. It does not refer to the logistical organisation of the demonstration. We believe the referee's concern follows from reading "organisational" in this second sense, and the requirement that the signature must precede the demonstration applies only under that reading, we apologize for the misunderstanding.

Once the claim is read as a statement about attention, the alternative the referee proposes is not in competition with it. Whether a user is physically present at the demonstration and posting from it, or following and engaging with the event remotely, both are directing sustained attention to the same situation, and both contribute to the same structural signature. This is precisely the phenomenon we characterise. Our analysis does not separate "organising" from "reacting", and it does not need to, because the object of study is the structure of concentrated attention, not the behaviour or timing of the physical event. For the same reason we prefer not to mark the times of the demonstrations in Fig. 2 and the analogous figures. The object of analysis is the temporal window over which attention concentrates, not the instant of the demonstration. Marking that instant would suggest a temporal correspondence we do not claim, and demonstrations have no sharply defined moment in any case, since some participants can arrive earlier and leave later.

To avoid this misreading by future readers, we have added a clarifying sentence to the section Discussion of the new version of the manuscript making explicit that "organisational principles of collective attention" refers to the self-organisation of attention in the network structure, and not to the logistical organisation of the demonstrations, and that we make no causal or predictive claim regarding the physical events.

- p. 7: The authors mention, “The presence of self-similarity indicates that local patterns of coordination and information diffusion are replicated at progressively larger scales, facilitating the rapid amplification and synchronisation necessary for successful mass mobilisation. This hierarchical self-organisation may represent an optimal structural configuration for balancing information dissemination with maintaining network coherence during critical periods.” This is overstated: self-similarity alone does not imply rapid information spread or amplification. Rather, it implies that the growth and development of these hashtag–hashtag networks might be governed by an underlying mechanism shared across scales. Conversely, if the authors argue that rapid dissemination of information is indeed key to successful mass mobilization, they should examine whether these networks are small-world or navigable during critical times, which is more relevant to information spread. In addition, “optimal structure configuration” is far beyond what the data suggests; it should be either accompanied by direct evidence, references (to make the statement general), or trimmed to avoid overclaiming.

Manuel Test de embeddings: Fuerza y suceso.

- Fig. 2: how are the widths of the yellow windows determined? How were those “minimum  $\epsilon_c^2(k)$ ”

values” identified exactly? Why is unit window not indicated in the figure? (The axis ticks instead indicate 10- or 8-hour windows, which is significantly longer than unit time, either 1 or 2 hours.) In fact, the dashed lines do not seem to necessarily match the time windows in which the errors are smallest within yellow bands. The authors should provide sufficient details about the methodology and reconcile the discrepancies between the CTWs and  $\epsilon$ -minimizing windows.

- Fig. 3 and Fig. SI 3: why are there no  $\epsilon$  values reported for the clustering coefficient distributions, especially when they argue that both distributions collapse onto single master curves?

Hay que calcular el epsilon cuadrado para el clustering también.

- Fig. 4: it would be nice to have a figure (including projections and connection probability distributions) showing what the embedding would look like during non-CTWs in SI, paralleling Fig 3 and SI 3. Since there is no quantitative assessment of the embedding in Fig. 4, it is almost necessary to show the difference between successful and unsuccessful hyperbolic embedding qualitatively.

Se debe calcular los embedding en la ventana no crítica del SI.

- p. 1: “..., fostering the emergence of effective social distances.” What does “effective social distance” mean? The authors should clarify its intended meaning and which results they refer to.

We thank the Referee for drawing our attention to this passage and for helping us recognize that the wording was not accurate. What we mean is that the combination of self-similarity and high clustering provides strong evidence for an underlying metric space and therefore motivates the use of hyperbolic embeddings to construct a geometric representation of the hashtag co-occurrence networks, in which effective distances between hashtags can be defined. We have revised the text in the manuscript to make this point clear.

- Methods section: There are many missing details about the implementation of hyperbolic embedding. There is no mention of convergence criteria, optimization constraints and arguments, what values of  $\beta$  and  $\mu$  were used, to name a few. The methodological details provided are not sufficient for the reader to understand and assess the validity of the method.

The embedding method and the corresponding code are described in detail in Ref xxx. In the new version of the manuscript, we provide Table xxx with the values of inferred parameters  $\beta$  and  $\mu$  and we also provide the validation metrics for the quality of the embedding xxx in the Supporting Information. *this is for Manuel.*

- Did the authors also analyze the degree distributions of the CTW networks? Combined with self-similar properties, degree distributions could provide more compelling evidence of criticality during physical demonstrations.

Re: The degree distribution of the CTW networks is already reported: the  $k_T = 0$  curve in Fig. 3a–c corresponds precisely to the (unrenormalised) degree distribution of the filtered CTW network for each dataset. The subsequent curves ( $k_T > 0$ ) show how this distribution transforms under the DTR procedure, and their collapse provides the self-similarity evidence requested by the referee.

- Methods Sec. B: which binning is used, logarithmic or linear? How was  $n_{\text{bin}} = 20$  selected? The fit can depend heavily on the number of bins. It should also be adjusted according to sample size.

Re: We thank the referee for the comment. Exponential binning has been used, as now stated explicitly in Methods Sec. C. The number of bins,  $n_{\text{bin}}=20$ , follows standard practice for this

type of curve-collapse analysis and was found to work consistently across all three datasets, allowing a single fit without the need for dataset-specific adjustment.

- Fig SI 3: how were the non-critical time windows chosen? Why?

Pensando en ajustar otra figura. Quizás una misma figura con el epsilon cuadrado de la otra medida (o reportar solo el peor epsilon).

## Minor comments

- The paper does not include information about data collection, e.g., how they obtained the data from X, whether it was accessed through a paid license, etc.

Re: We have created a new subsection into the section “Methods”, entitled Data Gathering, which reads: Twitter/X data for the Argentine events (No al Tarifazo and 9N) were collected by a collaborator through the Twitter/X Academic Research API, as part of a separate published study [6]. Data for the Charlie Hebdo event were collected independently, via the same academic API but at a different time, and were subsequently acquired commercially for this study.

- The caption of Fig. 2 is highly inconsistent and incomplete: (a–c) are incorrectly referenced. (c–i) are not even mentioned. The nestedness parameter is illustrated by green lines, not pink. There are two notations for  $\epsilon$  values,  $\epsilon_{cco}$  and  $\epsilon_c^2$ , which is confusing. The vertical dash-dotted lines are green, not black. Please correct all these inconsistencies and make sure the caption and main text are aligned in style and notation.

Re: All corrected. Thank you very much!!

Otra vez aqui se menciona los dos epsilon diferentes en figura 2. Y las lineas segmentadas, dice que son verdes en vez de negras, pienso que quizo decir gris. Manuel qué color son?

- Fig. 3 and main text: stick with either  $k_t$  or  $k_T$  throughout.

Re: Corrected. Thank you for pointing this out.

- The notation “ $\mathbb{S}^1/\mathbb{H}^2$ ” is confusing; the slash usually suggests a quotient space.

Re: We thank the Referee for this comment. We have long used the notation  $\mathbb{S}^1/\mathbb{H}^2$  to denote the isomorphism between the gravity-law formulation in  $\mathbb{S}^1$  and the hyperbolic formulation  $\mathbb{H}^2$  of the geometric model, and we therefore prefer to retain it.

- Methods Sec. A: what is  $g_i$ ? Please define it.

Re: We also thank the referee for this spot;  $g_i$  is the community assigned to node i, and it is now well defined in the manuscript.

- I suggest the authors might want to consider using the Savitzky–Golay filter to analyze the time-series data since it is a generalization of moving average and is often more effective.

Idealmente intentar y si no hay mucho cambio se argumenta para dejar el que tenemos ahora mismo.